

Homework 1: Exploratory Data Analysis Exemplar

IE 2064 Data Science

Dataset choice

We will use the mpg dataset. Note the mpg dataset is not a dataset in Chapter 1.4 but I decided to use it for the example answers as I do not want to distribute answers that could facilitate cheating in future years.

Data set fields description

Outcomes:

- **drv** (continuous) = City miles per gallon.
- **hwy** (continuous) = Highway miles per gallon

Predictors:

- **manufacturer** (categorical) = Name of the manufacturer, taking the values:

```
## [1] "audi"      "chevrolet" "dodge"      "ford"      "honda"
## [6] "hyundai"   "jeep"       "land rover" "lincoln"   "mercury"
## [11] "nissan"    "pontiac"    "subaru"     "toyota"    "volkswagen"
```

- **Model** (categorical) = Model of the car. Examples of some values include:

```
## [1] "a4"          "a4 quattro"   "a6 quattro"
## [4] "c1500 suburban 2wd" "corvette"     "k1500 tahoe 4wd"
```

There are a total of

```
## [1] 38
```

values.

- **disp** (continuous) = engine displacement, in litres. “Engine displacement is the measure of the cylinder volume swept by all of the pistons of a piston engine, excluding the combustion chambers” [Source: “Piston Engine Displacement”. The Engineering Toolbox. Retrieved 18 August 2021]. A summary of this column is:

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1.600   2.400   3.300   3.472   4.600   7.000
```

- **year** (continuous) = year the car was manufactured. A summary of this column is:

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    1.600   2.400   3.300   3.472   4.600   7.000
```

- **cyl** (continuous) = Number of cylinders in the engine. A summary of this column is:

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    4.000   4.000   6.000   5.889   8.000   8.000
```

- **trans** (categorical) = Type of transmission taking values:

```
## [1] "p" "r" "e" "d" "c"
```

- **drv** (categorical) = The type of drive train, where f = front-wheel drive, r = rear wheel drive, 4 = 4wd

```
## [1] "p" "r" "e" "d" "c"
```

- **fl** (categorical) = The fuel type taking values:

```
## [1] "p" "r" "e" "d" "c"
```

where p=premium, d=diesel, e=ethanol, r=regular, c=compressed natural gas - **class** (categorical) = “type” of car, taking values:

```
## [1] "compact"      "midsize"      "suv"          "2seater"      "minivan"
## [6] "pickup"       "subcompact"
```

CoNVO statement

Toyota builds cars. Fuel efficiency of their cars is extremely important. Research and development (R&D) at Toyota is trying to understand which new car models on the market have new technology that improves fuel efficiency. They use this information to make strategic decisions such as which technology to invest in.

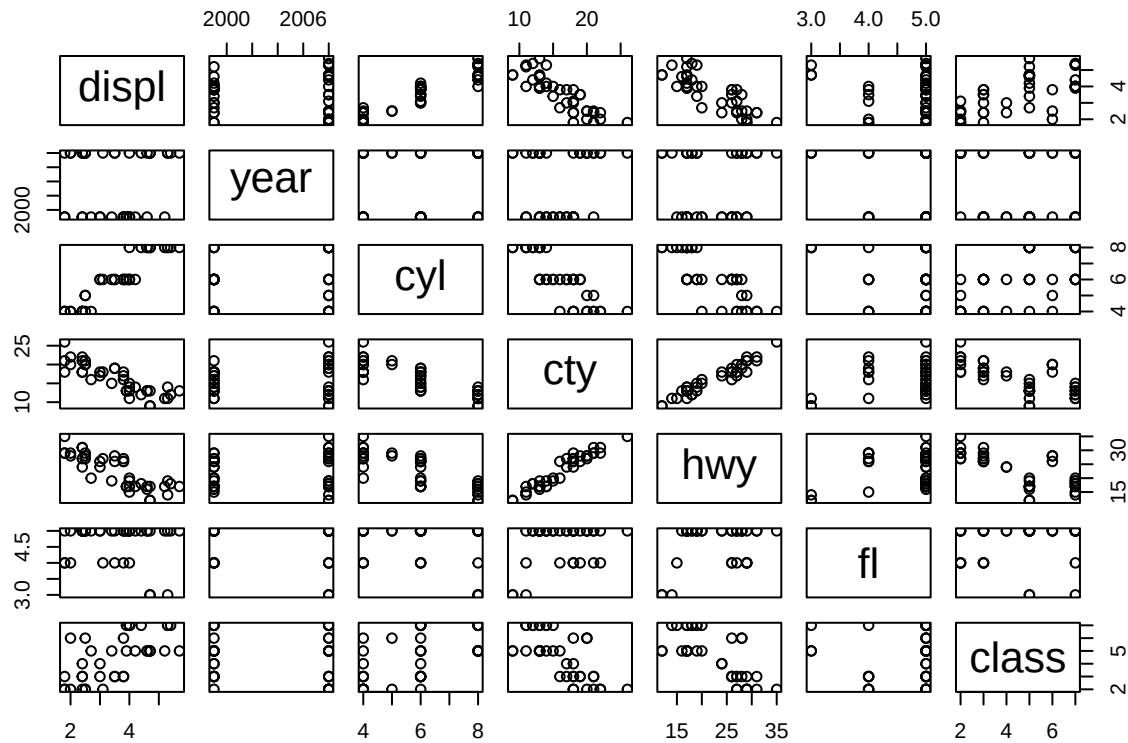
There are many new cars that enter the market each year. Toyota wants an indicator of which new cars have the best fuel efficiency technology, so they can do in-depth investigation of these cars. For example, by taking the car apart and inspecting its components. The difficulty is that one cannot consider only the cars with best raw fuel efficiency because cars with certain attributes are naturally more fuel efficient. For example: smaller cars tend to be more fuel efficient than larger cars; cars with smaller engines tend to be more efficient and so on.

The project will use data on historical cars (such as the mpg dataset) to build a predictive model on the fuel efficiency of the car. If a new car has much better fuel efficiency than predicted then it suggests the car includes new technology that gives improved fuel efficiency. R&D can therefore perform further investigation on that car.

If the head of R&D is happy with the preliminary results, a list of cars that are outperforming their predictions will be provided as a report to the head of R&D each month and she will use it to help discover new breakthroughs. In one year, the company will do a review to check if this tool has increased the speed Toyota discovers its competitors important technological breakthroughs.

Pairs plot

Pairs plot with all numeric variables and the two categorical variables (fuel type and type of car) that we suspect are critical. Fuel type is important because different fuels naturally have different energy storage capabilities and therefore different efficiencies. The type of the car is important because it tells us how big the car is which likely impacts fuel efficiency of the car.



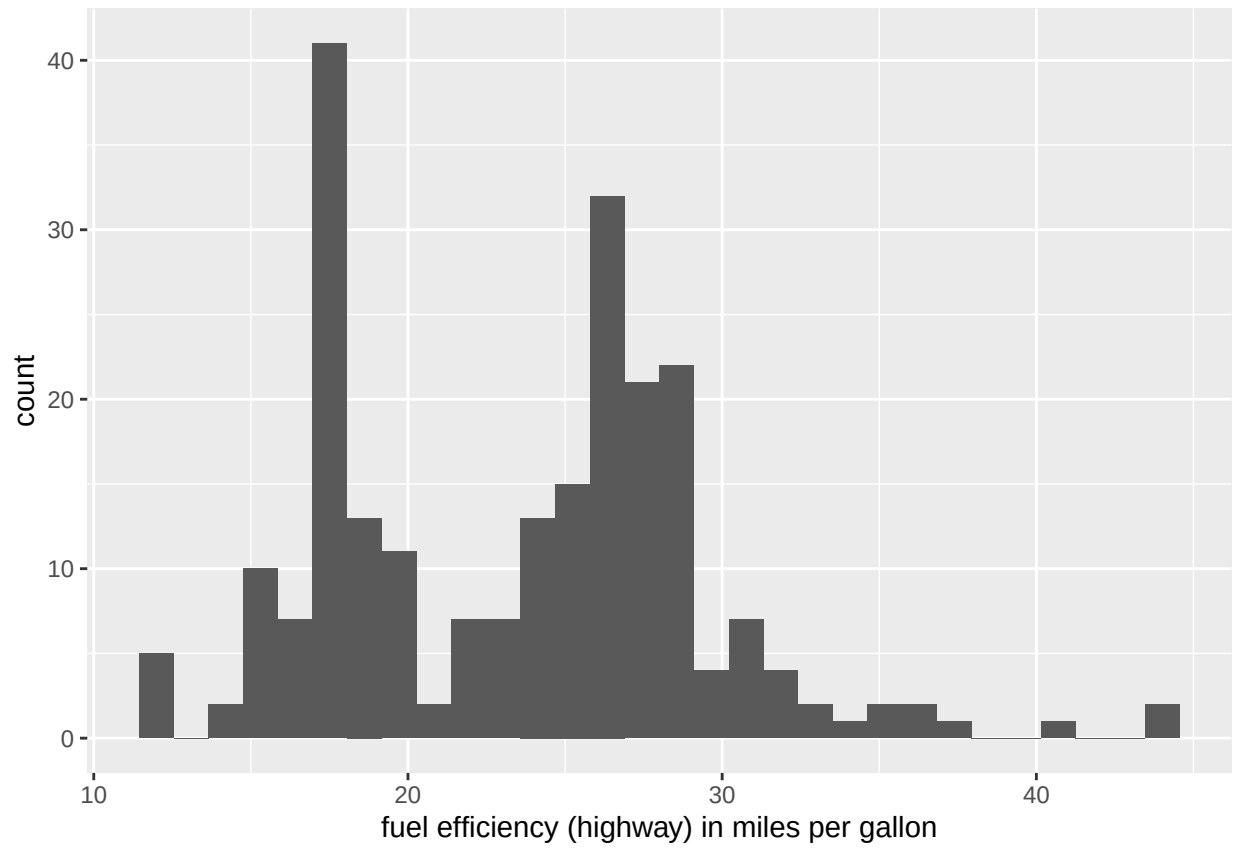
Identify an outcome and three predictor variables

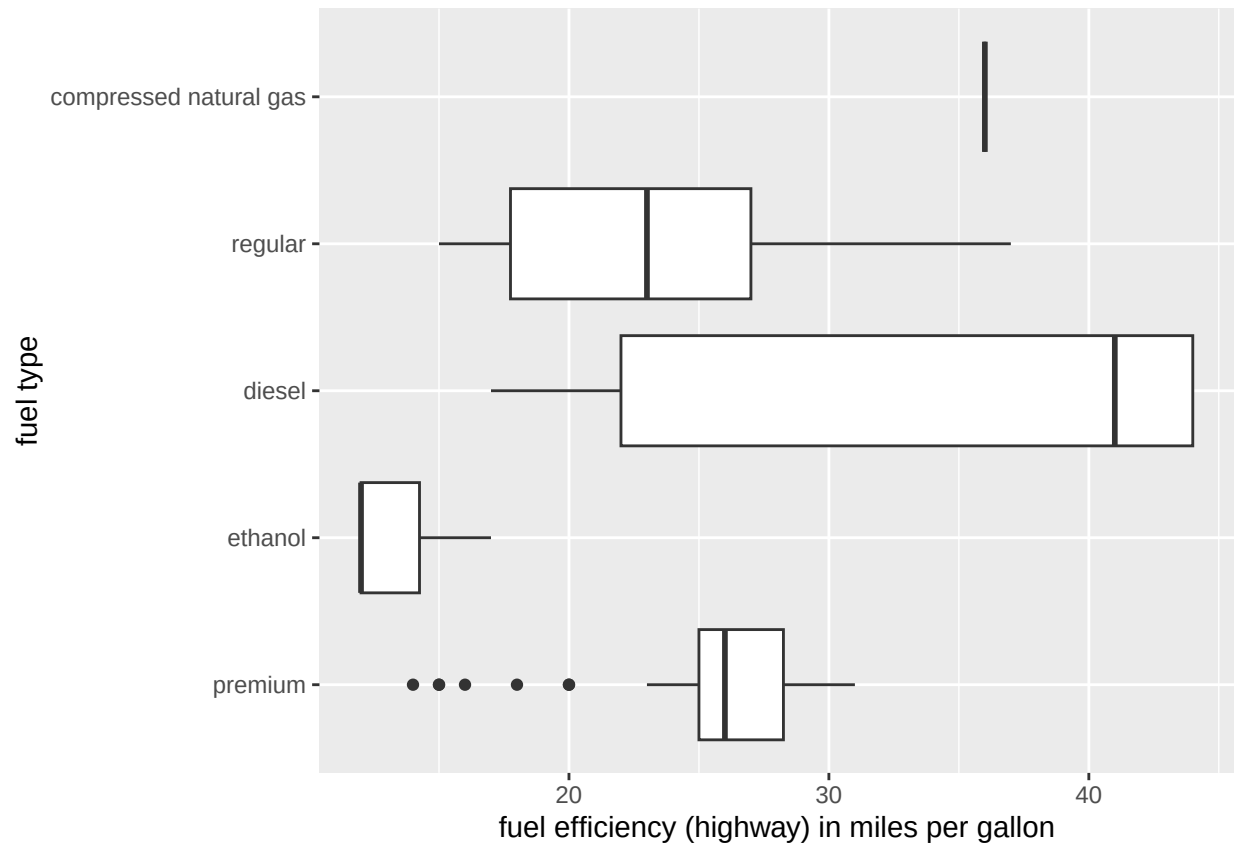
Note from the pairs plot that the cty and hwy are highly correlated. Therefore we will just focus on hwy as our outcome.

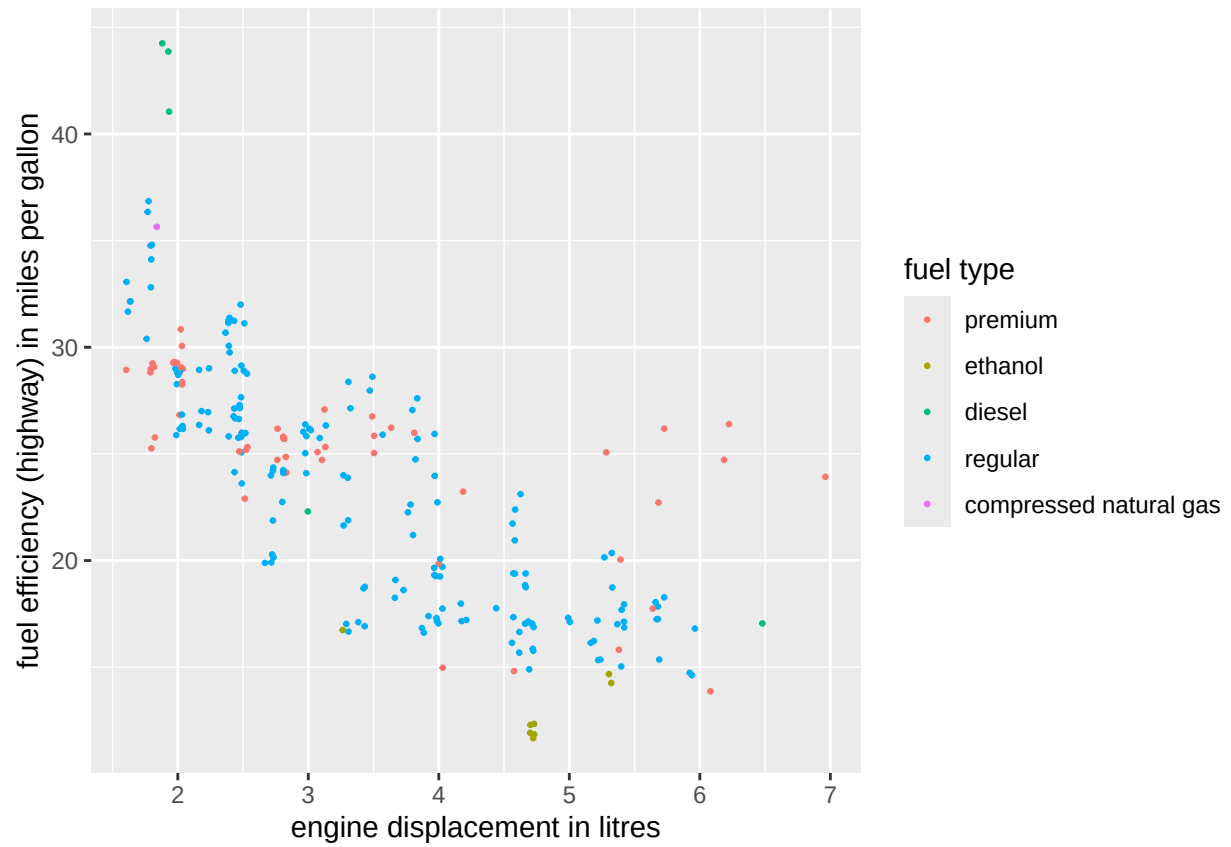
For the predictor variables we choose displ, fl and year because they appear uncorrelated with other predictors while having strong correlation with the outcome.

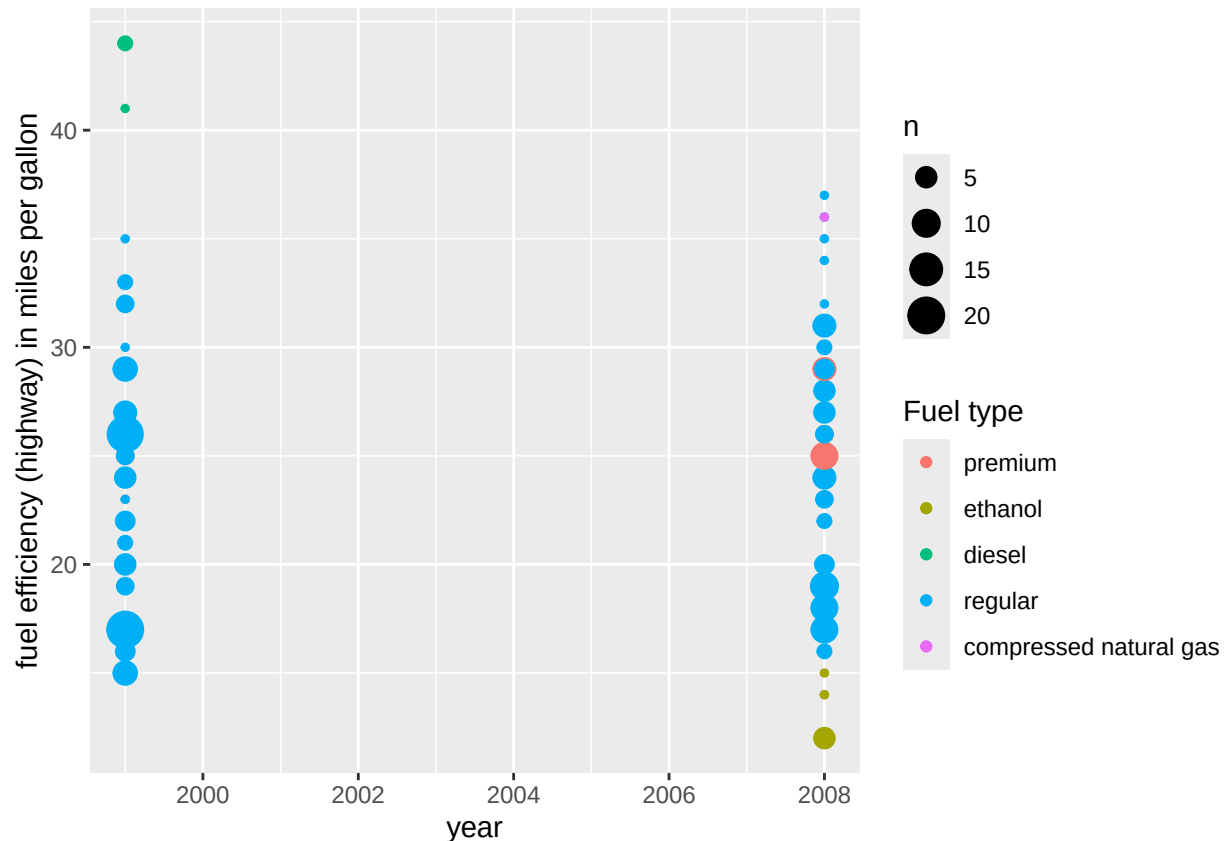
Explore the variables by applying five plot types

```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```









Causal relationships

From the scatter plot comparing engine displacement, fuel type and fuel efficiency we can see there is a strong negative relationship between engine displacement and fuel efficiency. We believe that increasing displacement is causing lower fuel efficiency due to the following statement (source: https://en.wikipedia.org/wiki/Engine_displacement)

“Engine displacement is the measure of the cylinder volume swept by all of the pistons of a piston engine, excluding the combustion chambers.[1] It is commonly used as an expression of an engine’s size, and by extension as a loose indicator of the power an engine might be capable of producing and the amount of fuel it should be expected to consume. For this reason displacement is one of the measures often used in advertising, as well as regulating, motor vehicles.”

We also note that from this scatter plot that there appears to be some correlation between fuel type and fuel efficiency. Here we also believe that fuel type is causing changes in fuel efficiency because different fuels (diesel, petrol, ethanol, etc) have different energy densities (https://en.wikipedia.org/wiki/Energy_density).

There are some plausible alternative hypothesis. For example, a potential confounding variable is customer preferences. For simplicity of this discussion lets restrict ourselves to the fuel types regular and premium fuels, these are both types of petrol and therefore have similar energy densities. Nonetheless we observe from our scatter plot that premium petrol is more fuel efficient than regular, particularly at large engine displacements. One possibility here is that wealthy customers demand high efficiency technology **and** premium fuel creating a spurious relationship.